



REPORT

Groundwater occurrence and quality in a hyper-arid desert: Insights from high-resolution electrical resistivity surveys in Tharparkar, Pakistan

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Abstract

The Tharparkar District in Pakistan is among the world's most water-scarce regions, characterized by its hyper-arid climatic conditions and the absence of perennial surface water resources. Understanding the occurrence of groundwater potential and its quality is therefore critical, particularly in remote regions where hydrogeological information remains limited. This study characterized groundwater resources using 576 high-resolution electrical resistivity survey (ERS) probes up to a depth of 300 m, conducted from 2017 to 2020. Complementary hydrochemical investigations included laboratory analysis of 168 groundwater samples collected from dug wells and 672 soil samples up to a depth of 0.60 m. Results demonstrated that the water table lies at depths of 2–15 m (26% of the area), 16–30 m (23%), and 31–90 m (51%). Interpretation of ERS data revealed that approximately 10% of the groundwater resources, covering the southern parts of Diplo, Islamkot, and Nagarparkar Talukas, are suitable for drinking at a depth of 25 m, with around 3% classified as fresh and 7% as slightly saline. Groundwater salinity increases with depth, as 51% of the area shows high salinity at depths of 26–50 m and very high salinity at depths of 51–200 m. Arsenic and fluoride concentrations exceed permissible limits in 10 and 28% of the groundwater samples, respectively. These findings provide robust geophysical and hydrochemical evidence to guide targeted groundwater development, water quality mitigation strategies and inform policy interventions, aimed at enhancing water security and safeguarding the health of communities living in severely water-stressed desert regions.

Keywords Arid region · Groundwater · Geophysical method · Arsenic · Pakistan

Introduction

Availability and access to good quality water is a fundamental requirement for all forms of life, including humans, and is essential for a healthy and balanced ecosystem (Benjmel et al. 2020). Beyond its biological importance, water is a critical resource for sustainable socio-economic development. Ensuring reliable access to water for domestic, agricultural, and industrial uses remains a major global challenge, particularly in areas with limited water resource availability and low and erratic rainfall (Abdelkareem et al. 2012; Prabhakar and Tiwari 2015). As the demand for freshwater continues to rise, global water shortages are projected to reach around 40% by 2030 (Achu et al. 2020). In this context, fresh groundwater represents a strategic reserve, especially in areas with arid and hyper-arid climatic and environmental conditions and where surface water is either scarce or not available. According to UNESCO (2015),

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groundwater supports around 2.5 billion people worldwide for domestic consumption, agricultural uses, and industrial activities (Mukherjee and Singh 2020).

Groundwater plays a critical role in sustaining life in desert ecosystems across the globe. In regions where surface water is virtually nonexistent and rainfall is low and highly variable over space and time, aquifers often represent the only dependable source of water for human, livestock, and agricultural uses (Smith et al. 2016). However, exploration, development, and sustainable management of groundwater in desert environments pose significant inherent challenges due to the complex subsurface geology, limited recharge, and increasing anthropogenic activities (Braune and Xu 2009; Smith et al. 2016).

Across the world's major deserts, including the Sahara in Africa, the Arabian Desert in the Middle East, the Kalahari in Southern Africa, the Sonoran Desert in North America, and the Great Victoria Desert in Australia, innovative and region-specific strategies have been adopted to address groundwater scarcity (Alao et al. 2024). These include the application of remote sensing and geophysical methods for aquifer mapping (Dailey et al. 2015; Elbeih 2015), managed aquifer recharge (MAR) systems (Aloui et al. 2023; Ajur and Baalousha 2021; Al-Muraikhi and Shamrukh 2017), artificial recharge basins (Abu-Taleb 2003; Gebreslassie et al. 2025), solar-powered deep boreholes, and community-level water governance frameworks (Alao et al. 2024). For instance, countries like Saudi Arabia (Al-Salamah et al. 2011; Masoud et al. 2018) and the United Arab Emirates (Salem et al. 2022; Eggleston and Mack 2020) have invested extensively in advanced hydrogeological surveys and groundwater modeling, while Namibia and Botswana have adopted decentralized water management and conservation techniques suited to rural desert populations (Arntzen 2003; GoB 2013). Despite differences in socioeconomic conditions, a common factor across successful desert groundwater management practices is the integration of scientific data with community-centered approaches and adaptive policy frameworks.

In comparison, the Thar Desert, located in southeastern Pakistan (Fig. 1), remains one of the most water-insecure deserts globally. With no perennial rivers or formal irrigation infrastructure, the population is largely rural and socioeconomically vulnerable, depending exclusively on groundwater extracted from dug wells and deep wells. Annual average rainfall is less than 200 mm and is highly variable both in magnitude and its spatial distribution. The aquifers are often shallow, discontinuous, and poorly recharged, with increasing evidence of salinization and localized overexploitation. Previous hydrochemical investigations conducted in the area revealed widespread groundwater quality concerns. Several studies (Ploethner 1992; Danishwar et al. 1995; Geyh and Ploethner 1995; Rafique et al. 2008; Brahman et al. 2013)

have reported elevated levels of electrical conductivity (EC), fluoride, and arsenic concentrations exceeding WHO permissible limits for drinking water with a significant proportion of samples classified as saline to highly saline. For instance, Alvi et al. (2008) analyzed 33 samples from Chachro, Mithi, and Diplo, reporting fluoride concentrations above the WHO guideline (1.5 mg/l) in 82% of samples and EC exceeding 2500 $\mu\text{S}/\text{m}$ in 79%, with values reaching 20,560 $\mu\text{S}/\text{m}$. Khuhawar et al. (2019) evaluated 2193 dug-well samples, classifying 27% as fresh to slightly saline, 15% saline, and 58% highly saline; arsenic and fluoride exceeded WHO limits in 29% and 36% of samples, respectively. Suthar et al. (2019) reported that only 26.4% of 53 samples from Mithi and Islamkot were suitable for both drinking and irrigation purposes, while 73.6% were saline to highly saline ($\text{EC} > 2600 \mu\text{S}/\text{m}$). Kumar et al. (2022) observed uniformly saline groundwater ($\text{EC} 2970\text{--}4200 \mu\text{S}/\text{cm}$) in Islamkot, with elevated concentrations of fluoride (1.01–2.02 mg/l) and arsenic (7–19 $\mu\text{g}/\text{l}$). Similarly, Kumar et al. (2022) found 52% of 25 samples from Mithi with high concentrations of fluoride (1.51–5.1 mg/l) and 36% with arsenic (15–80 $\mu\text{g}/\text{l}$), with 72% classified as saline to highly saline ($\text{EC} 2500\text{--}17,000 \mu\text{S}/\text{cm}$). More recently, Samtio et al. (2022) reported unsafe salinity levels ($\text{EC} 2593\text{--}18,950 \mu\text{S}/\text{cm}$) in 32 samples from Chachro. However, these studies mainly focused on the collection of groundwater samples from shallow dug wells and lacked integrated hydrogeological mapping of available groundwater potential. Consequently, the spatial distribution, variability with depth and aquifer development potential remain poorly understood. The lack of systematic geophysical investigation and groundwater monitoring has constrained evidence-based decision-making and sustainable water resource planning.

In this context, non-invasive geophysical techniques, especially Electrical Resistivity Surveys (ERS), have emerged as effective tools for groundwater investigation in arid and hyper-arid environments globally (Imran et al. 2023; Alao and Abubakar 2025). The ERS enables robust, cost-effective, and environmentally sound characterization of subsurface lithology, identification of aquifer zones, estimation of groundwater depths, and quantification of groundwater potential. The method is widely applied due to its affordability, operational simplicity, and high resolution (Telford et al. 1990; Helaly 2017). Furthermore, it provides valuable insights into porosity, permeability, and salinity variations when interpreted with hydrochemical data (Stewart 1982; Van Overmeeren 1989; Dahlin et al. 1999; Nowroozi et al. 1999; Meju 2005). When integrated with Geographic Information Systems (GIS) and field-based measurements, it significantly improves the accuracy of groundwater resource mapping and quality assessment in data-scarce areas.

This study leveraged high-resolution ERS in combination with water-table measurements, groundwater quality

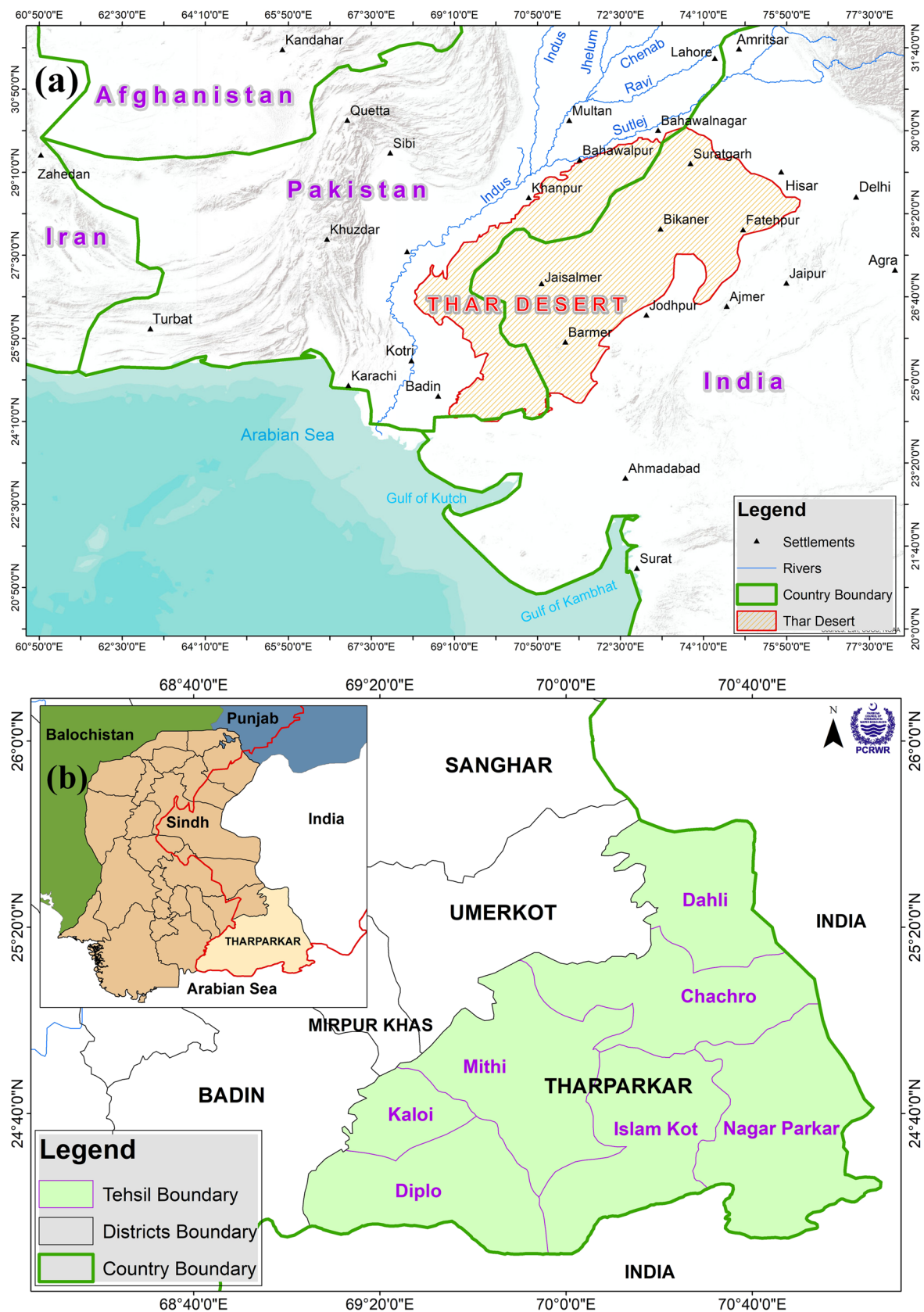


Fig. 1 a) Regional extent of the Thar Desert across Pakistan and India, and b) location of the Tharparkar District (study area) in Sindh, Pakistan

analysis, and soil texture characterization of the subsurface aquifer system across the Thar Desert with the aim to 1) delineate the spatial extent and distribution of groundwater aquifers, 2) evaluate groundwater potential and salinity patterns, and 3) assess water quality in relation to drinking water suitability. Through systematic characterizing, this research links local resource needs with global water security priorities, offering evidence-based strategies for managing water scarcity in arid landscapes.

This study advances groundwater research in arid environments by providing a regional-scale, high-resolution assessment of aquifer systems across the Thar Desert, an area where hydrogeological information has traditionally been sparse and fragmented. A key novelty lies in the integration of geophysical (ERS) data with groundwater quality parameters, allowing for the delineation of aquifer zones in relation to salinity distribution and drinking water suitability. This combined approach provides a more robust understanding of subsurface heterogeneity, linking lithological controls with hydrochemical processes. The methodological framework presented, combining high-resolution ERS, hydrochemical analysis, and GIS-based spatial mapping, offers a replicable, cost-effective approach for groundwater exploration in data-scarce, hyper-arid environments, with direct relevance to similar hyper-arid regions.

Materials and methods

Study area

The Tharparkar District extends between approximately 24–25°N latitude and 69–71°E longitude in southeastern Sindh Province, Pakistan, covering an area of 19,813 km². It is the largest district of Sindh province by land area. About 99% of it falls under the Thar Desert (Channa et al. 2021). Thar Desert is the 20th largest desert in the world, with a transboundary arid landscape shared between Pakistan (15% area) and India (85%). In Pakistan, it extends across the eastern part of Sindh province covering Tharparkar district and extends northward into southeastern Punjab province, where it merges with the Cholistan Desert. District Tharparkar is further divided into seven Tehsils (administrative boundaries also called Talukas); Diplo, Mithi, Nagarparkar, Chachro, Dahli, Kaloi and Islam Kot. Each Taluka functions as a local government subdivision, responsible for local governance and decentralized administration to facilitate effective public service delivery.

Climatic and hydrological conditions

The area experiences an extremely arid and highly variable climate that strongly influences its hydrology. Mean annual

maximum and minimum temperatures range from 34.7–36.9 and 17.5–19.9 °C, respectively, with May–June being the hottest and December–January the coldest months (Hanif and Ramey 2014). Over the 20th century, the region witnessed a temperature rise of 1.0–2.5 °C, accompanied by a 5–12 mm decline in annual rainfall and a 5–25% increase in humidity. Humidity peaks between June and September (59–71%), while strong winds and dust storms are common from May to July. Annual rainfall is highly erratic, ranging from 61 to 595 mm year⁻¹, with approximately 90% occurring during the monsoon months (June–September) (PMD 2014). Exceptional years, such as the 1361 mm recorded in 2011, contrast sharply with severe drought years including 2013, 2014, and 2018 (Usman and Nichol 2020). Winter rainfall is negligible (Hanif and Ramey 2014).

Hydrologically, the region depends almost entirely on this variable monsoon rainfall, which provides temporary surface water stored in small ponds or depressions lasting only a few months (Arisar 2011). Limited recharge leads to deep water tables (35–45 m) and a heavy reliance on groundwater that is often saline and contains high levels of arsenic and fluoride (Mangrio 2022; Kumar et al. 2022). Usable groundwater is mostly limited to shallow depths, particularly in the hilly Nagarparkar area where localized recharge occurs (Salam et al. 2023). Ongoing climatic and anthropogenic pressures continue to alter the groundwater regime, posing significant challenges to water security and livelihoods in this fragile desert environment (Mangrio 2022).

Geological and hydrogeological setting

The Thar Desert lies on the western margin of the stable Indian Peninsula forming a broader structural plain. This plain is largely covered by thick Quaternary sand dunes with an average height of around 80 m (Fig. 2). Rock outcrops are rare, except for limited granitic exposure in the Nagarparkar area, while scattered Mesozoic and Tertiary formations appear along the Indo-Pakistan border in the Jaisalmer and Rann of Kutch areas of India. Due to limited surface exposure, the understanding of the subsurface geology of the Thar Desert comes from geophysical surveys and drilling studies (Rehman et al. 1993; Fassett and Durrani 1994; Zaigham and Ahmed 1996). Seismic studies (Zaigham and Ahmed 1996) suggest that the Thar Desert lies above a granitic basement shaped by pre-Jurassic rifting that formed the Thar Basin (Fig. 3). The basement rises southeastward and deepens northwestward due to ancient tectonic movements (Zaigham and Ahmed 1996). Sedimentary sequences from the Mesozoic to the Tertiary indicate consistent depositional trends, reflecting the influence of early rifting.

Based on geoelectric information, drilling, and evidence from geological data, the region can be divided into four

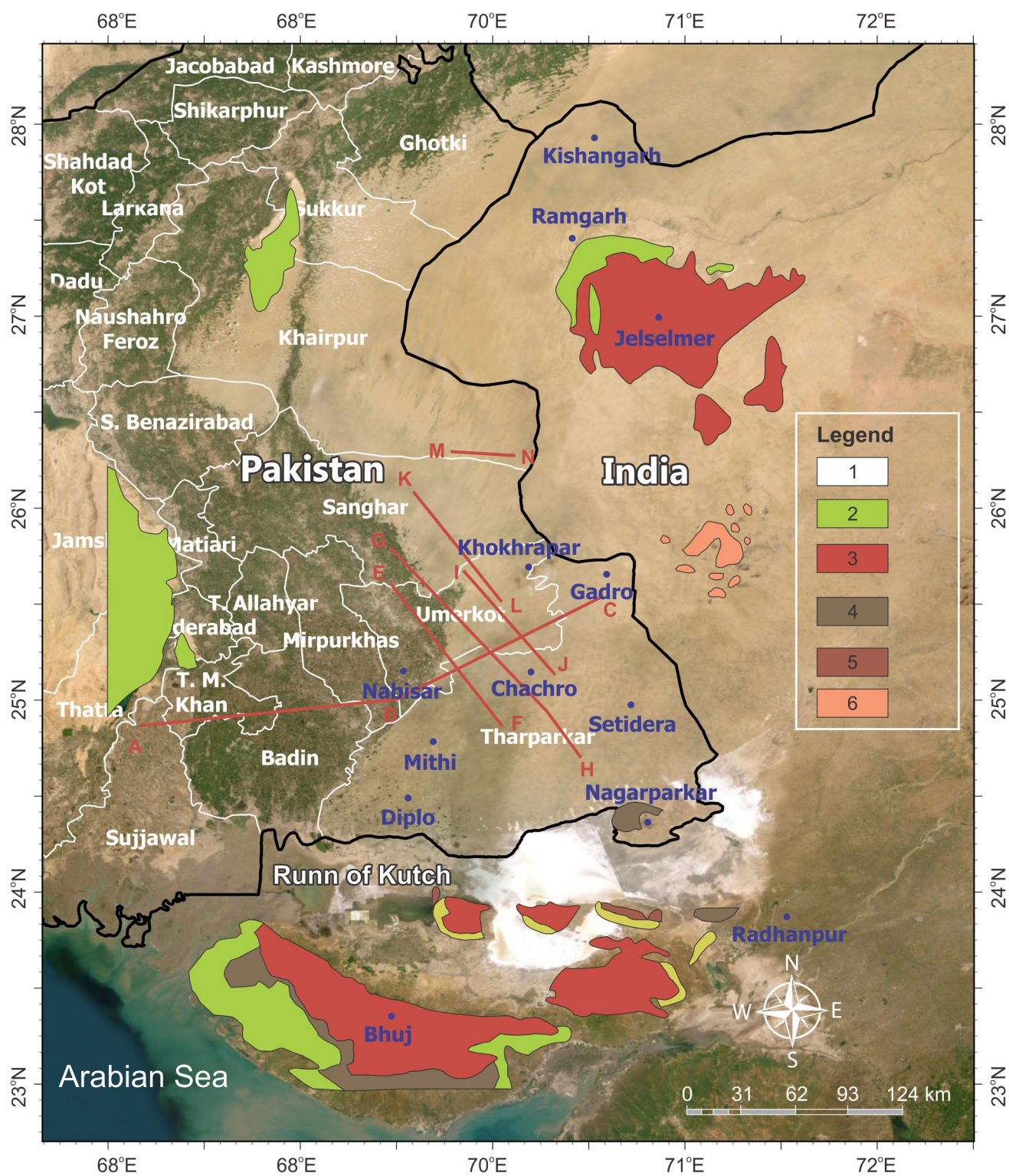
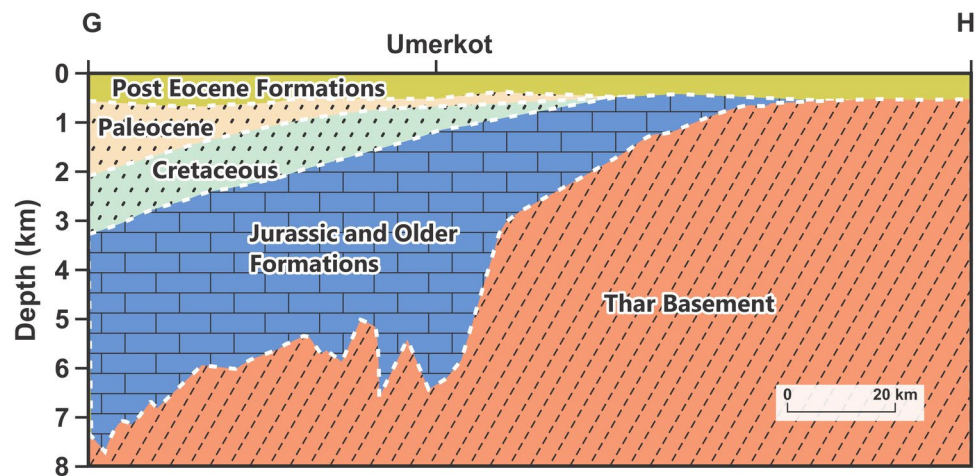


Fig. 2 General geology of the study area showing locations of seismic profiles and geological cross section. Geologic units: 1) Quaternary sediments/Desert dune-sands, 2) Tertiary sequences, 3) Mesozoic-Jurassic sequences, 4) Lava flows- Deccan traps, 5) Precambrian units, 6) Rhyolites-Malani beds

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Fig. 3 Geological cross section of the Thar Rift derived from seismic interpretation illustrating the structural configuration of the basin and underlying basement architecture (modified from Zaigham and Ahmed, 1996)



major lithological units (Rehman et al. 1993; Fassett and Durrani 1994; Zaigham and Ahmed 1996) (Fig. 4):

1. *Dune zone (loose zone)*: Composed of well-sorted sand dunes with 8–15% clay and silt ranging from 14–81-m-thick covering northeastern and southern areas (Chachro–Islamkot–Mithi). This zone forms the surficial deposits across much of the Thar Desert and hosts localized shallow aquifers.
2. *Oxidized zone*: Layers of clays, silts, and sands with ironstone concretions and siderite nodules, 38–107 m thick, largely of sub-recent age and is generally considered non-water-bearing.
3. *Coal-bearing formations*: Alternate bedding of claystones, siltstones, sandstones, and lignite with cumulative lignite up to 34 m. These layers form confined aquifers under artesian conditions.
4. *Basement complex*: Consists predominantly of weathered and fresh granite, with the weathered zone having varying thickness. The thickness of granite basement rock is relatively shallow (1–10 m) but varies spatially. The granitic basement is encountered mainly in the east and southeast of Chachro), with variations including rhyolitic, basaltic, and dioritic rocks near Pabban and the Indian border. It underlies the coal-bearing succession and forms the hydrogeological basement of the aquifer system.

The findings of the geoelectrical surveys (Schildknecht 1991) suggested that the high-resistivity (50–150 Ω m) basement complex represents a deeply fissured sandstone aquifer with potential for freshwater occurrence. Subsequent vertical electrical sounding (VES) investigations conducted south of Chachro refined this interpretation by identifying the basement as comprising massive granitic basement and layered Archean metasediments (Zaigham and Ahmed 2000). These studies indicate that groundwater potential is associated with

weathered and fractured zones developed within the crystalline basement rather than with a laterally continuous sandstone aquifer.

Hydrogeologically, the region relies entirely on groundwater due to the absence of perennial surface water. The inventories of dug wells spanning over approximately 8500 km² covering the areas between Gadro and Virawah (Ploethner 1992), indicate the existence of perched aquifers within sandy-silty layers at the interface of the dune and oxidized zone, sealed by clay beds. Water table depths (WTD) range from 5–15 m near Gadro to 30–45 m at Chachro, becoming deeper towards the west. Generally, dug wells extend to 20–80 m, tapping groundwater that is mostly saline to brackish. Confined aquifers within the coal-bearing formations provide moderate yields. Thin aquifer layers occur above Coal Seam A (Fig. 4) and between Seams B and C, while a thick sandstone aquifer (0–34 m) lies below Seam C (Fassett and Durrani 1994) offering variable water quality with electrical conductivity ranging from 4000–5000 μ S cm⁻¹. Recharge is limited and largely controlled by monsoon rainfall, but pumping tests indicate that sustained abstraction can enhance both water availability and groundwater quality in localized areas.

Soil and groundwater sampling and analysis

Soil samples were collected adjacent to dug wells using a 10 km $\times$ 10-km sampling grid. Overall, 672 soil samples were collected from four depth intervals: 0–15 cm, 15–30 cm, 30–45 cm, and 45–60 cm. The samples were analyzed to determine soil texture and electrical conductivity (EC). Soil texture was determined using the Bouyoucos Hydrometer method (Bouyoucos 1962), while EC was measured using the saturation paste extract method (USDA 1954). These analyses were conducted at the Soil and Water Analysis Laboratory of the Drainage and Reclamation Institute of Pakistan (DRIP), Tandojam, which is a

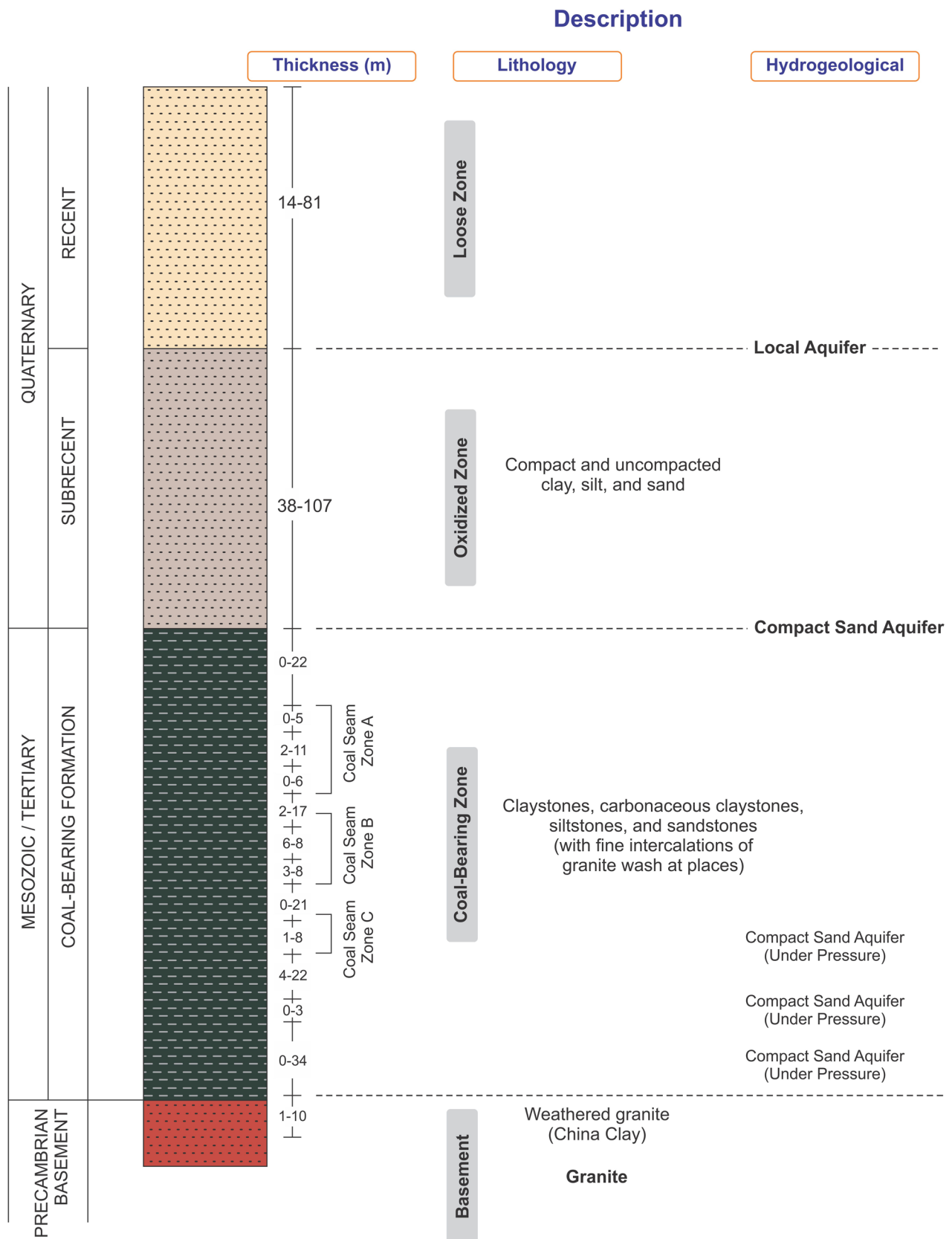


Fig. 4 Generalized lithological column of the Thar Desert, illustrating stratigraphic units encountered in coal exploration boreholes and their associated hydrogeological characteristics (modified after JTB 1994)

regional laboratory of Pakistan Council of Research in Water Resources (PCRWR), having accreditation (ISO-17025) in Pakistan.

Water table depth was measured using a water level indicator in the monitoring dug wells (Sheikh and Ansari 2019). In addition, 168 groundwater samples were collected from the same wells (Fig. 5) and analyzed for EC and selected heavy metals, including arsenic and fluoride. Electrical conductivity was measured using a calibrated Hanna 2210 EC meter (Jakhani et al. 2019). Arsenic concentration was determined using Arsenic Merck MQuant Test Kit (Reddy et al. 2020), while fluoride was analyzed using the SPADNS method (Rashid and Rizwan 2013). Spatial distribution maps of soil properties, groundwater quality, and water table depth were developed using ArcGIS software.

Electrical resistivity survey (ERS)

The ERS was employed as the primary technique for groundwater investigation. Based on Ohm's law, ERS measures subsurface resistivity variations associated with

lithology and groundwater characteristics. The survey was conducted using the ABEM Tetrameter SAS 4000 with a Schlumberger configuration. A total of 576 vertical electrical soundings were carried out at a grid of 5 km × 5 km, covering depths up to 300 m to ensure systematic spatial coverage (Fig. 5). Apparent resistivity data were processed using Interpex (IX1D) software to estimate true subsurface resistivities. These values were subsequently converted to groundwater EC using a regression relationship analysis developed with laboratory-measured EC data (Iqbal et al. 2020). The regression model, originally developed for the Lower Indus Plain (Sindh, Pakistan), was applied in this study to relate resistivity to EC (Fig. 6). The model shows a good level of agreement, with a coefficient of determination (R^2) of 0.78, indicating a satisfactory goodness-of-fit. Groundwater quality was classified into four categories (Iqbal et al. 2020): freshwater (< 1500 $\mu\text{S/m}$), slightly saline (1600–2500 $\mu\text{S/m}$), saline (2600–4000 $\mu\text{S/m}$) and highly saline (> 4000 $\mu\text{S/m}$). The ERS data were mapped using ArcGIS software by overlaying parameters at various depth intervals.

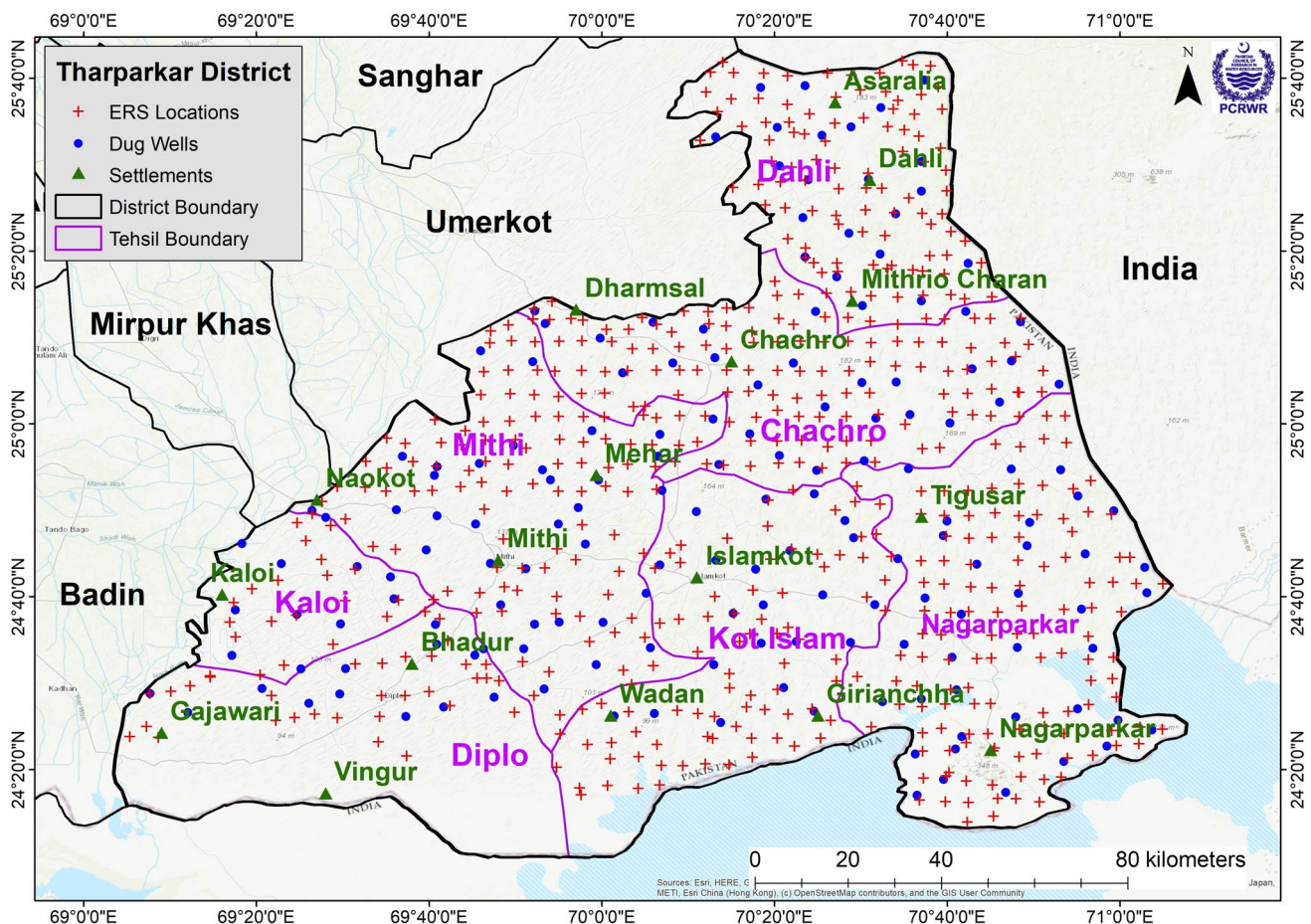
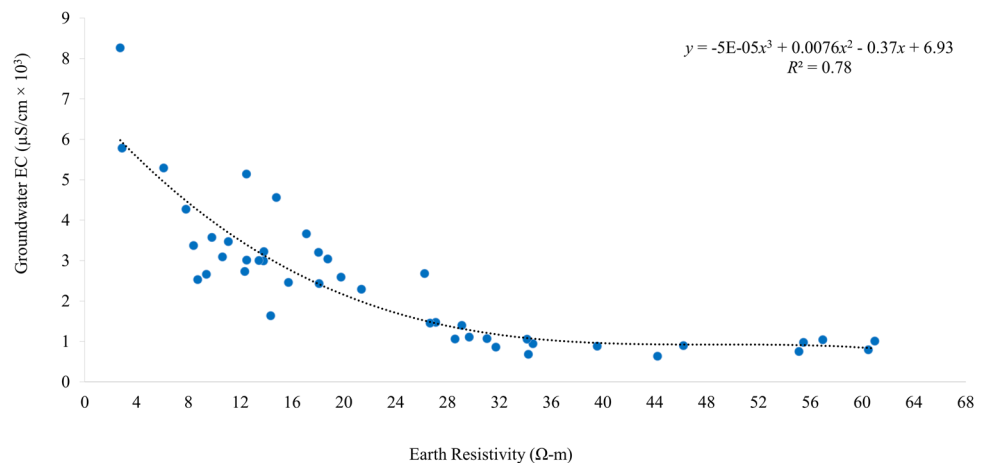


Fig. 5 Location of the dug wells and ERS in Tharparkar District

Fig. 6 Regression between earth resistivity and groundwater quality developed for lower Indus Plain area (adopted from Iqbal et al. 2020)



Electrode configuration and field measurements

The survey employed a Schlumberger array, in which four electrodes are aligned symmetrically along a straight line. The outer electrodes (A and B) inject current into the ground, while the inner electrodes (M and N) measure the resulting potential difference (Fig. 7a). In this configuration, AB represents the full current-electrode separation and AB/2 its half-spacing from the center; similarly, MN represents the full potential-electrode separation and MN/2 its half-spacing. During the surveys, AB/2 was progressively increased to increase investigation depth, while the spacing between potential electrodes (MN/2) was kept comparatively small and increased only when the measured potential difference became too weak for reliable recording.

During measurement, a controlled direct current (DC) was introduced into the ground through electrodes A and B and the resulting potential difference between M and N was recorded by the terameter (Fig. 7b). Electrode spacing was systematically increased to investigate deeper subsurface layers, while maintaining a fixed central point for each sounding. At each location, multiple readings were obtained corresponding to different depth intervals of the investigation.

Data processing and interpretation

Apparent resistivity values were plotted against electrode spacing to generate curves. The interpretation of resistivity data was carried out using computer-aided techniques using software 1X1D, from Interpex, USA. An iterative inversion technique was applied to estimate true resistivities and the thickness of subsurface layers. Model parameters were progressively refined during successive iterations to minimize the root mean square error (RMSE) between interpreted and observed data, achieving an error of less than 5%. The final model with the lowest error (<5%)

was selected, representing appropriate subsurface conditions. To enhance reliability, interpretation was guided by available geological and hydrogeological information to select the most realistic model. This integrated approach enhanced the robustness and reliability of the subsurface characterization, leading to improved groundwater assessment in the study area.

Results

Hydrogeological setting: Soil texture and recharge conditions

The spatial analysis of soil properties (Fig. 8a) indicates that sandy soils dominate across the region at all the investigated depths, with localized occurrences of sandy clay loam in parts of Chachro, particularly in the upper (0–30 cm) layers. These coarse-textured soils are characterized by high infiltration rates and low moisture retention capacity, which influence recharge dynamics in the region.

The soil salinity analysis based on electrical conductivity of the extract (ECe) (Fig. 8b) shows that most of the study area falls within the non-saline category ($\text{ECe} < 4000 \mu\text{S}/\text{m}$) across all depths, suggesting that soil salinity is not a primary limiting factor for recharge; rather, low rainfall, high evapotranspiration, and deep water tables likely restrict effective groundwater replenishment. Overall, the hydrogeological setting indicates that while infiltration potential is high, actual recharge to aquifers remains limited, particularly in areas with thick unsaturated zones. These findings highlight that despite favorable infiltration characteristics, natural recharge remains constrained, indicating the need for targeted interventions such as managed aquifer recharge and rainwater harvesting to enhance groundwater replenishment in the region.

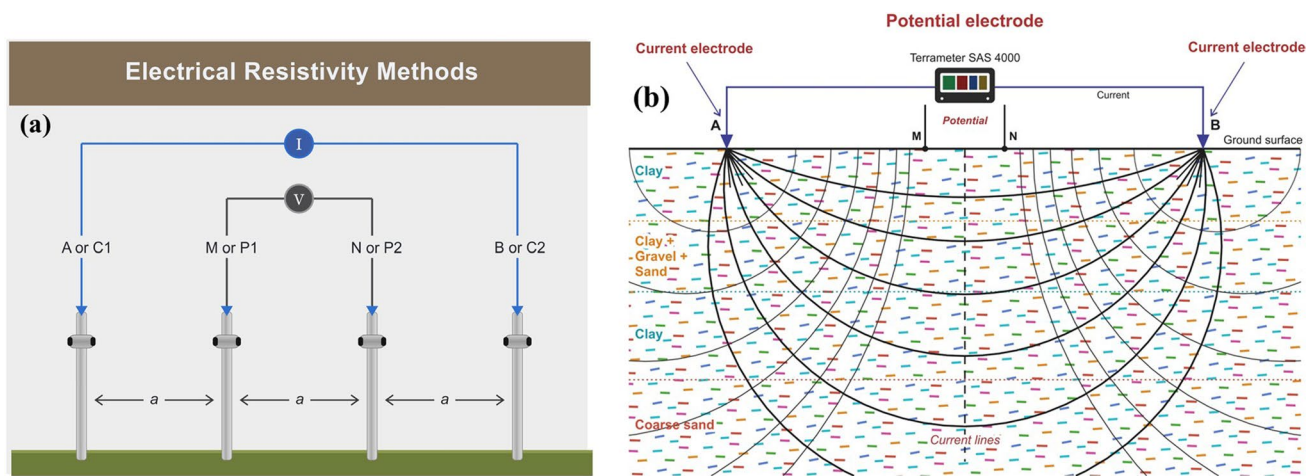


Fig. 7 a) Setting of potential and current electrodes, and b) operational working diagram of terrameter (Source: Dars et al. 2023)

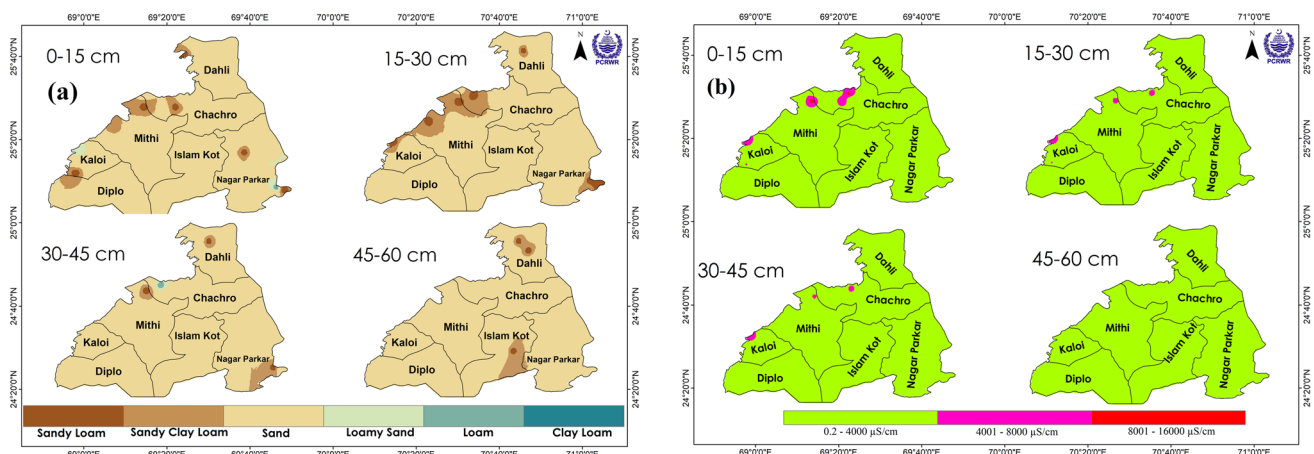


Fig. 8 a) Geospatial variation in the soil texture, and b) the soil EC at different depths in Tharparkar

Groundwater occurrence: Shallow vs. deep systems

Groundwater across the region exhibits strong spatial variability (Fig. 9a) with clear distinctions between shallow and deep groundwater systems. Shallow groundwater (≤ 15 m) is primarily confined to southern Talukas, including Diplo, Islamkot, Nagarparkar, and parts of Dahli. These areas represent localized recharge zones, characterized by low-lying topography and relatively favorable hydrological conditions. Nagarparkar shows consistently shallow water tables. In contrast, moderate to deep groundwater systems (≥ 30 m) dominate much of the district, particularly in Mithi and Chachro, where water table depths exceed 60 m. The spatial distribution (Fig. 9b) shows that 26% (5151 km^2) of the area contains shallow groundwater (2–15 m), 23% (4557 km^2) lies within moderate depths (16–30 m) and significant portions of the district exhibit deeper groundwater conditions (> 45 m).

These results demonstrate that groundwater availability is not only limited but also spatially fragmented, with shallow aquifers occurring as isolated pockets within a predominantly deep groundwater system. This spatial variability in groundwater occurrence underscores the importance of site-specific groundwater development strategies, where shallow aquifer zones may be prioritized for sustainable abstraction, while deeper zones require cautious management and recharge enhancement measures.

Groundwater quality

Salinity patterns and depth-wise trends

Groundwater quality across region is dominated by salinity, with a clear depth-dependent deterioration pattern, as shown in Figs. 10 and 11, 12, 13, 14 and 15. At shallow depths (≤ 25 m), groundwater occurrence and quality (Fig. 10)

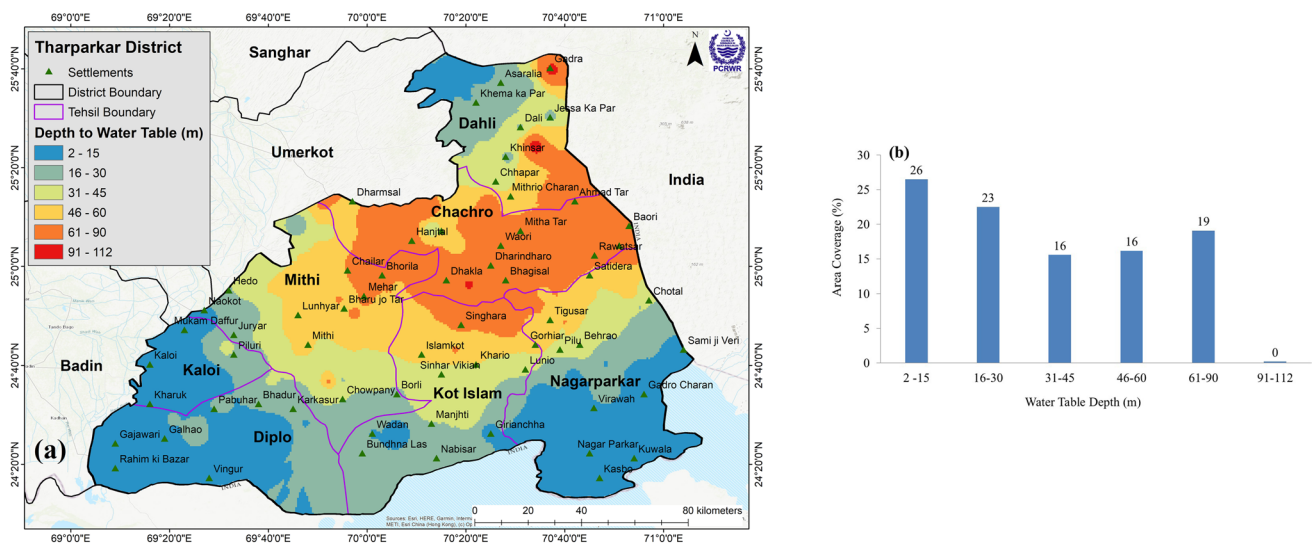


Fig. 9 a) Spatial variations of WTDs and b) area coverage under different WTDs

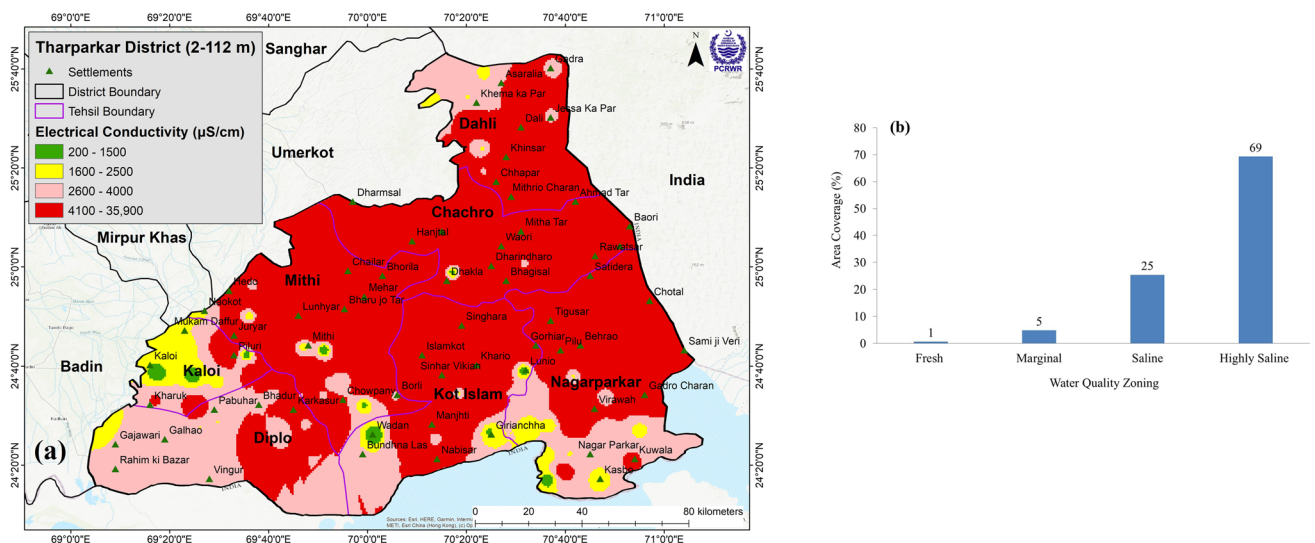


Fig. 10 a) Groundwater quality at variable depths (2–112 m) and b) area coverage in Tharparkar at variable depths (2–112 m)

indicate that the vadose (unsaturated) zone dominates 59% of the area, saline to highly saline groundwater occupies 31% and only limited pockets of fresh (3%) and slightly saline (7%) groundwater occur, primarily in southern Talukas. At intermediate depths (26–50 m), groundwater quality (Fig. 11) deteriorates further, with highly saline conditions expanding to 51% of the area, while freshwater zones decline to negligible levels.

At greater depths 51–75 m (Fig. 12), highly saline groundwater dominates 76% of the area, at 76–100 m (Fig. 13), this increases to 82% and at > 100 m (Fig. 14), the groundwater is almost entirely saline, reaching ~99% with

highly saline conditions. Overall, groundwater EC ranges from 4100 to 35,900 $\mu\text{S}/\text{m}$ across 69% of the study area (Fig. 15), reflecting widespread salinity. However, localized lenses of relatively lower salinity groundwater ($\text{EC} \leq 4000$ $\mu\text{S}/\text{m}$) occur in southern Talukas (Fig. 15), though these are generally unsuitable for drinking and may only support livestock or saline agriculture. These results highlight that groundwater quality is more severely constrained than groundwater availability, particularly with increasing depth.

This depth-dependent deterioration in groundwater quality has direct implications for water resource planning, indicating that shallow aquifers offer comparatively better

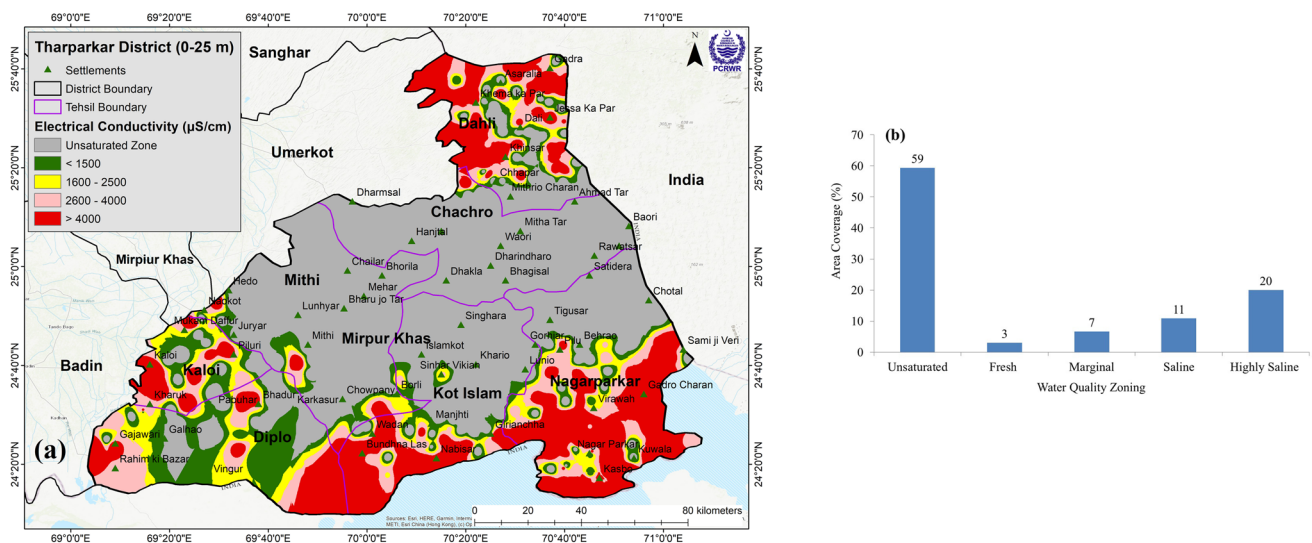


Fig. 11 a) Spatial variations of EC and b) area coverage of EC at 0–25 m depth in Tharparkar

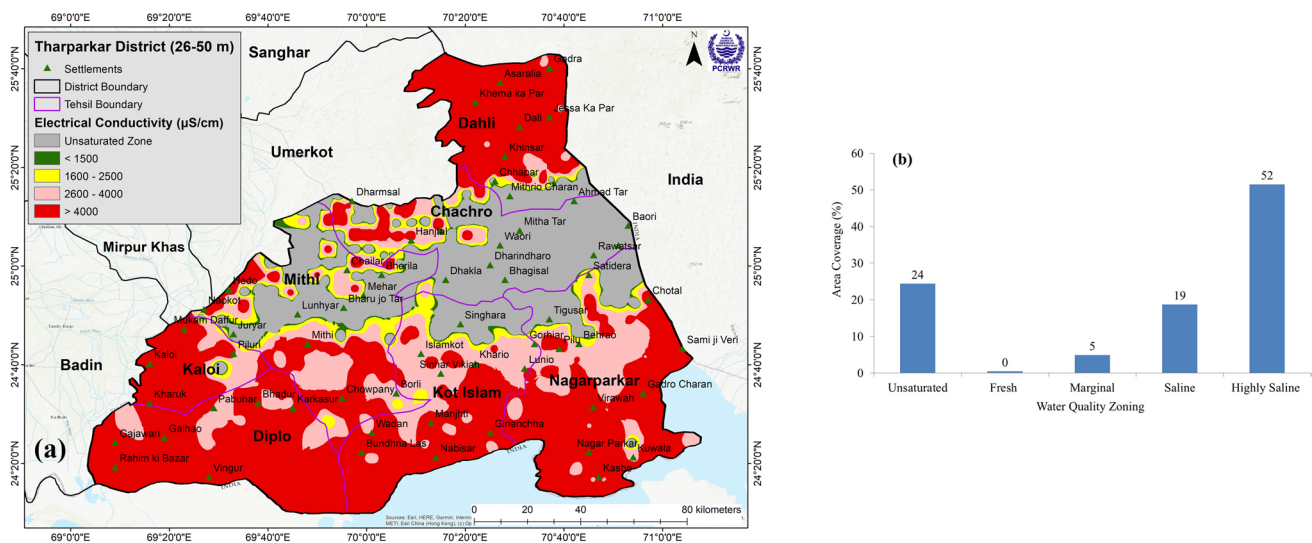


Fig. 12 a) Spatial variations of EC and b) area coverage of EC at 26–50 m depth in Tharparkar

opportunities for potable use, while deeper saline groundwater may require alternative utilization strategies.

Figure 12 shows the quality of groundwater at depths from 26 to 50 m. Groundwater in most areas was found to be highly saline at these depths, particularly in the southern part of its Talukas Diplo, Islamkot, Nagarparkar and Dahli. As the investigation depth increases from 25 to 50 m, the area occupied by freshwater-bearing aquifers decreases to 1% (198 km²), the area of unsaturated zone declines to 24%, and the area with highly saline groundwater expands to 51% (10,105 km²). At depths from 26 to 50 m, slightly saline and saline groundwater extended over 4755 km² (24% of the area surveyed).

Figure 13 shows the groundwater quality from 51 to 75 m depth. At these depths, the highly saline groundwater intensifies as discerned across 15,058 km² (76% of the entire study area), particularly within certain areas of Talukas Mithi, Chachro, Diplo and Nagarparkar. Unsaturated zone, with slightly saline, and saline groundwater extends across 4755 km², 24% of the area. Groundwater quality at depths between 76 and 100 m was identified as highly saline, within 82% (16,247 km²) of this zone occurring in Mithi, Chachro, Diplo and Nagarparkar, whereas the remaining 18% (3566 km²) falls in other saline zones (Fig. 13). Figure 14 illustrates the groundwater quality from 101 to 300-m depth. Groundwater

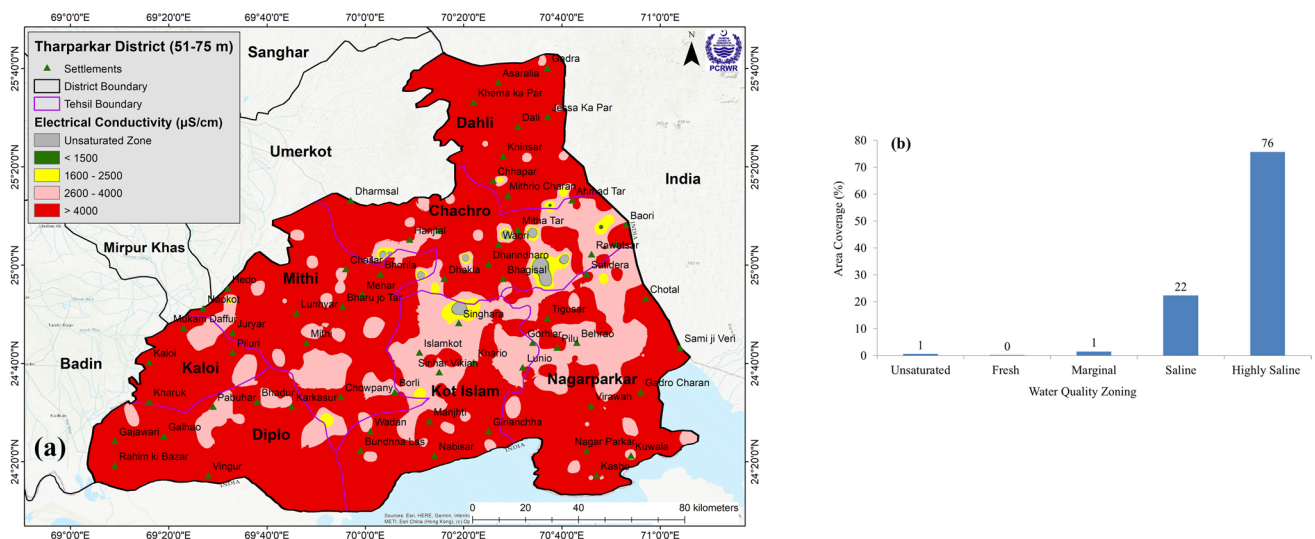


Fig. 13 a) Spatial variations of EC and b) area coverage of EC at 51–75 m depth

with a high salt concentration prevailed over 85% and 88% of the area at depths of 101–150 m and 151–200 m, respectively. Almost the entire groundwater domain (99%), covering an area of 19,615 km² was identified as highly saline.

Figure 16 shows the area coverage under different groundwater quality classes (fresh, slightly saline, saline, and highly saline) at each depth interval, which clearly demonstrates the depth-wise trend in groundwater salinity. Groundwater quality in the study area exhibits a strong depth-wise deterioration trend, with saline conditions increasingly dominating at greater depths. At shallow depths (≤ 25 m), fresh and slightly saline groundwater is limited to about 10% of the area (3% fresh and 7% slightly saline), while saline to highly saline water occupies 31%, and a substantial 59% remains within the unsaturated zone. At intermediate depths (26–50 m), groundwater quality declines markedly, with freshwater reduced to only about 1%, slightly saline and saline water together covering 24%, and highly saline groundwater expanding to 51% of the area. This trend intensifies further at 51–75 m, where highly saline groundwater dominates 76% of the area, leaving only about 24% under relatively lower salinity classes. At depths of 76–100 m, highly saline conditions increase to 82%, with the remaining 18% comprising saline categories. Beyond 100 m, groundwater is almost entirely saline, with highly saline water covering approximately 99% of the area. Overall, these patterns clearly demonstrate a progressive increase in salinity with depth, indicating that shallow aquifers offer comparatively better, though still limited, groundwater quality, whereas deeper aquifers are overwhelmingly constrained by high salinity.

Arsenic and fluoride

In addition to salinity, groundwater quality is affected by geochemical constituents, notably arsenic and fluoride, as shown in Fig. 17(a and b). Arsenic concentrations remain within safe limits (< 10 $\mu\text{g/l}$) in approximately 90% of the study area, while water from 10% of dug wells exceeds WHO guidelines, particularly in Chachro and parts of Mithi and Diplo (Fig. 17a).

Arsenic in groundwater across the Indus Basin is predominantly geogenic in origin, as consistently reported by Pakistan Council of Research in Water Resources (PCRWR) through its nationwide water quality monitoring programs. The PCRWR investigations, supported by large-scale sampling, have demonstrated that arsenic occurrence is widespread but spatially heterogeneous, with concentrations exceeding the WHO guideline (10 $\mu\text{g/l}$) in multiple regions of Punjab and Sindh (Rashid and Rizwan 2013).

These studies indicate that arsenic mobilization is primarily controlled by water–rock interaction processes within alluvial aquifer sediments, particularly through the reductive dissolution of iron oxyhydroxides, which releases adsorbed arsenic into groundwater. The Indus Basin aquifer system, composed largely of Holocene alluvial deposits, provides favorable geochemical conditions for this mobilization, especially in low-lying floodplain environments with longer groundwater residence times.

These assessments further highlight that arsenic concentration is not uniformly correlated with groundwater salinity or fluoride, reflecting the complex and site-specific hydro-geochemical controls governing its occurrence. In contrast to humid alluvial settings where arsenic is often associated with fresh groundwater, arid regions such as Tharparkar

exhibit limited recharge, high evaporation, and saline groundwater systems, which alter the dominant geochemical dynamics with reduced widespread arsenic mobilization.

Fluoride occurrence is more widespread with 28% of sampled wells exceeding the WHO limit of 1.5 mg/l (Fig. 17b), making water unsuitable for drinking. The remaining 72% of samples fall within acceptable limits. These findings indicate that groundwater quality challenges in the region are not only limited to salinity but also involve significant geochemical health risks, further constraining the usability of available water resources. Fluoride enrichment is influenced by alkaline conditions, prolonged water–rock interaction, and low calcium activity, which enhance the dissolution of fluoride-bearing minerals (Adimalla and Venkatayogi 2017).

These findings emphasize the need to integrate groundwater quality monitoring with public health planning, ensuring that safe drinking water interventions are prioritized in areas where arsenic and fluoride concentrations are high.

Discussion

The role of soil texture is vital in replenishing the groundwater through recharge from rainfall and seepage from irrigation networks. The dominant soil textures in the study area are sandy and sandy clay loam, which are categorized as coarse-textured soils. The moisture-retention capacity of these soils is typically low and they exhibit high infiltration and drainage rates compared to fine-textured soils (Bockheim 2023). Soil with a sandy clay loam texture was found in certain areas of Chachro Taluka in the upper (0–15 cm) and middle (15–30 cm) layers. This soil type enhances moisture retention, while the sandy component reduces compaction and improves drainage. High soil salinity levels can negatively impact crop growth and productivity. It may leach down to further deteriorate the groundwater quality at shallow depths (e.g., in dug wells). Soil salinity (EC) assessment suggests that the soil profile is predominantly non-saline and generally conducive to healthy plant growth.

Overall, the dominance of coarse-textured sandy soils suggests that while infiltration potential is high, effective groundwater recharge remains limited due to low rainfall, high evapotranspiration, and deep water tables. This indicates that soil properties alone do not control recharge; rather, climatic and hydrogeological conditions play a more decisive role, resulting in restricted replenishment of aquifers across much of the study area.

Shallow WTDs (2–15 m) on 26% of the total surveyed area were found in southern Diplo, Islamkot, Nagarparkar, and northern Dalhi. The shallow WTDs in these areas are probably due to their lowland characteristics and closeness to the Arabian Sea. Moreover, a tributary originating from the Naukot branch of the Nara canal is used for irrigated

agriculture in Dahli Taluka. The seepage from this tributary and irrigation return flows might have led to a rise in the water table in this Taluka. The WTDs are mostly shallow in Nagarparkar, which can be attributed to the seepage from 36 small dams constructed by the Government of Sindh and the influence of Rann of Kutch marsh. Rann Kutch is a saline dried salt lake in the dry season and a flooded sea during the monsoon season (Ali 2009). Deeper WTDs (61–90 m) in 19% of the total surveyed area were found in central Chachro and the northern part of Mithi Taluka. Mithi and Chachro Talukas show significantly deeper WTDs than Nagarparkar and Diplo, most probably due to the upland characteristics and lack of any dams or perennial streams (Khuhawar et al. 2019; Suthar et al. 2019). Despite sandy soil strata and an abundance of rainfall in monsoon months, with an average of 376 mm, the WTD is deeper in Chachro and Mithi. Generally, sandy soil promotes groundwater recharge because of its high infiltration rates. However, the presence of a deep water table or thick vadose zone (Walker et al. 2020), together with high evapotranspiration and the absence of any perennial streams (Khuhawar et al. 2019; Geyh and Ploethner 1995), prevents the infiltrated water from reaching the aquifer (Mangrio 2022; Khuhawar et al. 2019; Geyh and Ploethner 1995).

In summary, groundwater occurrence in Tharparkar is highly uneven and strongly controlled by local hydrogeomorphic conditions. Shallow water tables exist in isolated pockets due to localized recharge, whereas deeper groundwater dominates most of the region due to limited recharge and thick vadose zones. This spatial variability highlights that groundwater availability is both constrained and fragmented, requiring location-specific water development and management approaches. The groundwater quality in Taluka Chachro is predominantly saline, most likely due to limited recharge associated with low rainfall conditions, relatively deeper water table depths, and the influence of an oxidized zone. The oxidized zone is composed of compact to loose clay, silt, and sand. Groundwater in the region is generally saline, reflecting the geological and paleoenvironmental history of these areas. Most of the Indus Basin, including parts of the Thar Desert, was historically influenced by marine transgressions associated with the Arabian Sea. As the sea gradually retreated, saline deposits and marine signatures remained embedded within the sedimentary formations, contributing to the inherent salinity of groundwater (Ashraf et al. 2012; Mundorff et al., 1976; Ploethner 1992; Naseem and Rafi 2024). In addition, geochemical processes such as mineral weathering and water–rock interaction further control groundwater salinity in the Indus Plain.

Over time, in irrigated regions of the Indus Plain, recharge from irrigation networks and rainfall has led to the development of a relatively thin freshwater layer over the saline groundwater, which is widely utilized for domestic

and agricultural purposes. In contrast, in the desert areas including Tharparkar District, the rainfall is limited and the irrigation network is nonexistent, resulting in very low or no recharge. Consequently, groundwater chemistry in these regions remains largely controlled by lithology and low recharge conditions, resulting in predominantly saline groundwater.

However, localized pockets of fresh to slightly saline quality groundwater occur where shallow water table conditions, relatively higher recharge, and favorable lithological settings prevail. In arid environments such as Thar, limited recharge combined with high evaporation further concentrates dissolved salts, reinforcing the naturally saline character of the groundwater. Spatial analysis indicates that groundwater with EC ranging from 4100 to 35,900 $\mu\text{S}/\text{m}$ covers approximately 69% of the Tharparkar area, reflecting widespread salinity. These results align with earlier studies in the region. Alvi et al. (2008) reported EC exceeded 2500 $\mu\text{S}/\text{m}$ in 79% of samples with maximum values reaching 20,560 $\mu\text{S}/\text{m}$. Khuhawar et al. (2019) based on 2193 dug-well samples, classified 73% as saline to highly saline. Similarly, Suthar et al. (2019) found that 73.6% of samples from Mithi and Islamkot areas were unsuitable for drinking and irrigation purposes due to high salinity. Similarly, Kumar et al. (2022) reported that 72% of 25 samples from Mithi were saline to highly saline (EC 2500–17,000 $\mu\text{S}/\text{m}$) while Samtio et al. (2022) observed EC values ranging from 2600 to 18,900 $\mu\text{S}/\text{m}$ in Chachro, indicating unsafe salinity levels.

The southern part of Talukas Diplo, Mithi and Nagarparkar exhibits localized lenses of comparatively lower salinity groundwater (EC $\leq$ 4000 $\mu\text{S}/\text{m}$) at varying depths. Although not suitable for drinking, such water may be utilized for livestock and saline agriculture. Only a limited fraction of groundwater, 1% fresh and 5% slightly saline quality, with EC ranging from 180 to 2500 $\mu\text{S}/\text{m}$, occurs in isolated pockets within Talukas Diplo, Mithi and Nagarparkar. These zones represent potential sources for drinking water purposes.

Overall, groundwater salinity in the area reflects a combination of geological inheritance and ongoing hydrogeochemical processes, with a clear trend of quality deterioration with increasing depth. This depth-dependent increase in salinity is driven by limited recharge, high evaporation, and prolonged water–rock interactions. The occurrence of fresh to slightly saline groundwater in localized shallow zones indicates that recharge and favorable lithological conditions can partially moderate salinity; however, such zones remain limited.

Anthropogenic and geological formations are determining factors of arsenic concentration in groundwater resources, leading to severe life-threatening effects if used for drinking for longer periods (Brahman et al. 2013; Salam et al. 2023). The adverse impacts of arsenic on human health have been

reported globally. The gastrointestinal tract absorbs arsenic when such groundwater is used for drinking for extended periods, resulting in many diseases such as cardiovascular disease, diabetes, cancer, and mental health issues (e.g., Bhowmick et al. 2018; Salam et al. 2023). As per the WHO Guidelines, an arsenic level of greater than 10 $\mu\text{g}/\text{l}$ is unsafe for drinking (Bibi et al. 2015). Khuhawar et al. (2019) found that the concentration of arsenic in 29% of groundwater samples exceeded the limit set by WHO, whereas in the current study, 10% of the surveyed dug wells showed a higher concentration than 10 $\mu\text{g}/\text{l}$ of arsenic. This difference may be attributed to the fact that groundwater samples were collected from 168 dug wells in the current study, while Khuhawar et al. (2019) collected water samples from 2193 dug wells. A larger sample size can influence the overall results by affecting the sampling distribution. Due to the large sampling size, Kumar et al. (2022) found 36% of 25 samples from Mithi with arsenic level of 15–80 $\mu\text{g}/\text{l}$.

Fluoride is an essential micronutrient for maintaining human health, and at concentrations below 1 mg/l, it plays a key role in strengthening the apatite matrix of skeletal tissues and teeth (Meenakshi, 2006). However, a fluoride concentration of greater than 1.5 mg/l in drinking water is unsafe for drinking (Khuhawar et al. 2019). In the current study, the concentration of fluoride was found to be less than 1.5 mg/l in 72% of the surveyed dug wells while 28% of the dug wells exceeded the safe limit. However, Khuhawar et al. (2019) indicated a higher percentage of fluoride, with 36% of groundwater samples exceeding the safe limit. This difference can also be associated with a bigger sample size. They also reported 52% of 25 samples from Mithi with high levels of fluoride (1.51–5.1 mg/l).

The presence of arsenic and fluoride further complicates groundwater quality in the area, reflecting geogenic controls associated with aquifer lithology and hydrogeochemical processes. Although their occurrence is spatially variable, these contaminants indicate that groundwater quality challenges extend beyond salinity alone, underscoring the need for integrated water quality management and targeted monitoring.

Groundwater investigated through ERS at varying depths was interpreted based on EC. The ERS survey showed that at a WTD of 25 m, groundwater was available in 41% of the area (3% fresh, 7% slightly saline, and 31% saline), while the vadose zone predominated, covering the remaining 59% area. There is no or little groundwater recharge in most parts of the region. Installing recharge wells in areas with deep water tables to harness rainfall-induced runoff could be an effective approach to augment groundwater recharge and increase the fresh groundwater lenses. Pakistan Council of Research in Water Resources (PCRWR) has developed a technology for groundwater recharge through rainwater harvesting (Arshad et al. 2023). Hundreds of recharge wells have been installed in Punjab and Balochistan provinces and

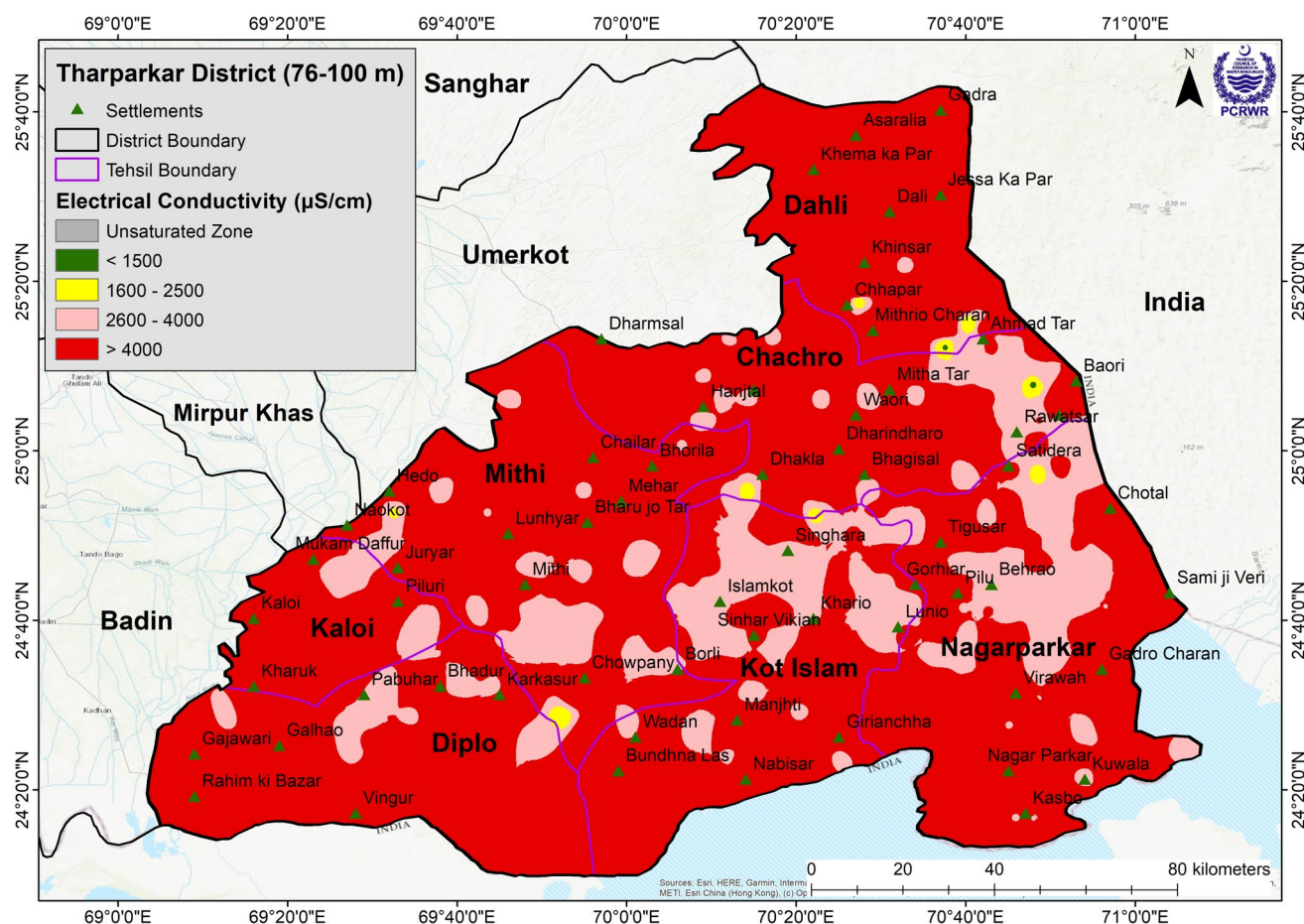


Fig. 14 Spatial variations in EC at 76–100 m depth

the Federal Capital Islamabad. The analysis of water quality from these wells indicated that the use of filter pits at recharge wells has resulted in a reduction of turbidity from 80 to 6 nephelometric turbidity units (NTU), which, in turn, has led to an improvement in groundwater quality, especially in total dissolved solids (TDS) (Arshad et al. 2023). Approximately 55% of the total precipitation received during the 2021 monsoon period contributed to groundwater recharge through injection wells. In Thar, to reduce the cost of drilling, a dug well can be converted to a recharge well to serve the same function, harnessing the potential of groundwater aquifer recharge.

Groundwater in most areas was found to be highly saline at depths between 26 and 50 m, particularly in the southern part of its Talukas Diplo, Islamkot, Nagarparkar and Dahli. Within this depth range, freshwater-bearing aquifers decrease to 1%, the unsaturated zone declines to 24%, slightly saline and saline water zones reduce to 24%, and areas with highly saline groundwater expand to 51% of the surveyed region. From 51 to 75 m depth, the highly saline groundwater intensifies, as discerned across 76% of the entire area, within certain areas of Talukas Mithi, Chachro,

Diplo and Nagarparkar. Khuhawar et al. (2019) reported that about 58% of the population uses saline groundwater for drinking with EC exceeding 4680 $\mu\text{S}/\text{m}$ in the areas of Mithi, Chachro, and Diplo Talukas. Their findings confirm the results of the current study, where a substantial deterioration in groundwater quality is found, where the EC exceeds 4000 $\mu\text{S}/\text{m}$ in 69% of the area in Mithi, Chachro, and Diplo Talukas. Moreover, Khuhawar et al. (2019) reported that in these Talukas, 15% of the population consumes groundwater with EC values between 2340 and 4680 $\mu\text{S}/\text{m}$, which is considered relatively less harmful, while 27% relies on groundwater with EC below 2340 $\mu\text{S}/\text{m}$, suitable for consumption without many adverse effects. However, Natesh et al. (2020) found that the consumption of low-quality groundwater has resulted in serious health issues for the communities living in the region. Therefore, the use of such groundwater should be discontinued. An important option is to harvest the available rainfall using various techniques (Imran et al. 2023). These hydrogeochemical conditions suggest that conventional groundwater development for drinking purposes is limited, and adaptive strategies such as selective abstraction of shallow aquifers

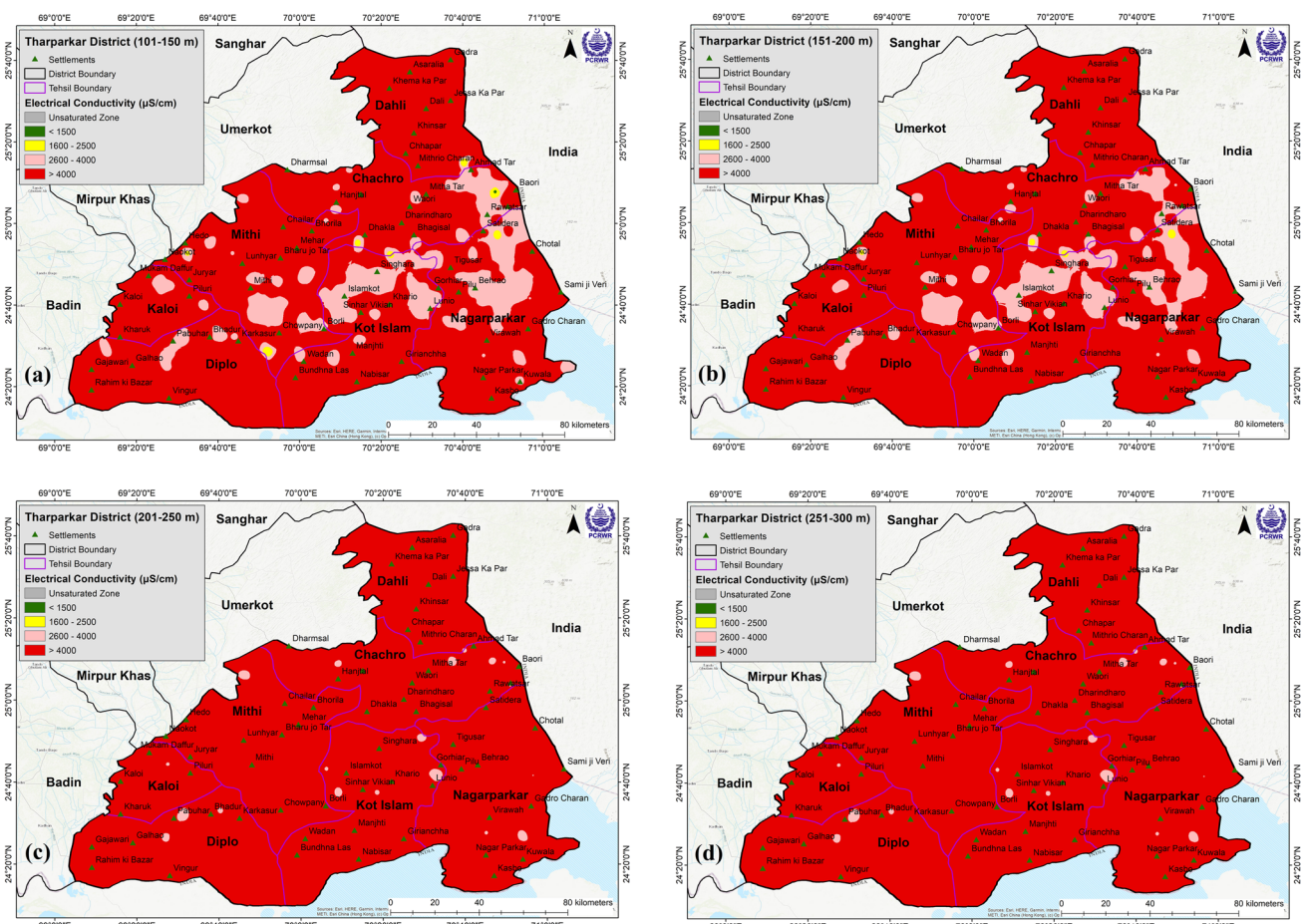


Fig. 15 Spatial variations in EC at a) 101–150 m, b) 151–200 m, c) 201–250 m, and d) 251–300 m WTDs

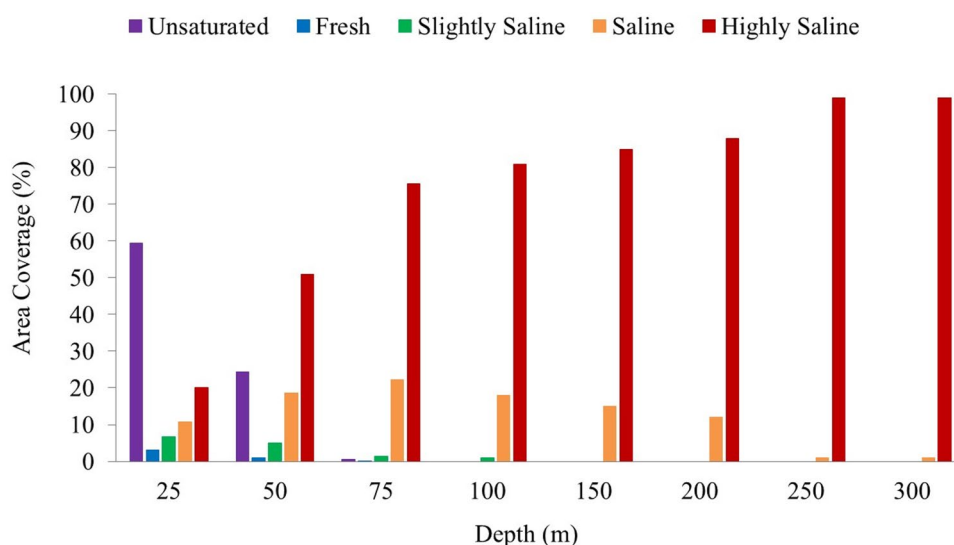
and utilization of saline groundwater for agriculture and aquaculture become increasingly relevant.

The widespread lenses of saline groundwater (< 4000 $\mu\text{S/m}$) could be utilized for saline agriculture and aquaculture practices. Salt-tolerant cereal crops and vegetation types, such as grasses, shrubs, and trees, are potential options for saline agriculture (Qureshi and Perry 2021). Under these circumstances, groundwater with high salinity could be used productively. However, the sustainability of saline agriculture is restricted by salt accumulation in the soil profile, which is widely dependent upon the salinity level of irrigation water. No salt accumulation occurs in a well-drained soil irrigated with irrigation water with salinity up to 6250 $\mu\text{S/m}$ in a maize cropping system (Feng et al. 2017). A comprehensive and recent study by Gul et al. (2023) on saline aquaculture in Mithi demonstrated a good potential of different salt-resistant fish species stocked in saline water of 9000 $\mu\text{S/m}$. Therefore, adapting saline agriculture and aquaculture practices using saline groundwater could be a potential and alternative livelihood option.

The results presented above provide a comprehensive understanding of groundwater occurrence and quality across the Tharparkar District. Translating these hydrogeological insights into actionable strategies is essential for improving water resource management in this hyper-arid environment.

The hydrochemical characteristics of groundwater in the Thar Desert suggest that groundwater quality is controlled by the combined influence of arid climatic conditions, lithology, aquifer depth, and water–rock interaction processes. Limited recharge associated with low and erratic rainfall, high evapotranspiration, and prolonged groundwater residence time contributes to progressive salinity increase with depth (Khuawar et al. 2019). The geology of the Thar Desert further influences groundwater chemistry, where the aquifer system mainly comprises sand dunes, oxidized clay–silt sequences, coal-bearing sedimentary formations, and granitic basement rocks. Moreover, the groundwater in the region is predominantly Na–Cl type and becomes increasingly saline with depth. Shallow aquifers, particularly in southern Talukas, contain relatively lower salinity groundwater due to localized recharge

Fig. 16 Summary of the area coverage under different quality classes at each depth



and shorter groundwater flow paths, whereas intermediate and deeper aquifers exhibit increasingly saline conditions.

Arsenic occurrence appears to be predominantly geogenic, associated with sedimentary aquifer materials and water–rock interaction processes involving iron-bearing minerals. In contrast, fluoride enrichment is likely associated with alkaline groundwater conditions, long hydrological residence time, and dissolution of fluoride-bearing minerals within aquifer sediments and granitic basement-derived materials. A positive association exists between fluoride, sodium, bicarbonate, salinity, and alkaline groundwater conditions (pH 7.1–8.6) under predominantly Na–Cl type groundwater settings (Khuhawar et al. 2019). Such hydrochemical conditions favor fluoride dissolution and stabilization in groundwater. The presence of granitic basement rocks and associated fluoride-bearing minerals in the southeastern parts of Tharparkar, particularly near Nagarparkar, may further contribute to elevated fluoride concentrations through weathering and mineral dissolution processes.

Overall, groundwater quality in the Thar Desert exhibits a clear depth-dependent hydrogeochemical pattern, characterized by increasing salinity and fluoride concentrations with depth, while arsenic occurrence remains comparatively localized and spatially variable. These findings provide a first-order regional hydrogeochemical and qualitative interpretation of groundwater quality processes derived from integrated geophysical and hydrochemical observations.

Study limitations and future prospects

This study provides a regional-scale assessment of groundwater occurrence and quality using high-resolution ERSs integrated with hydrochemical data. However, it has certain limitations that need to be addressed in future studies. The conversion of resistivity to EC was based on an existing regression model developed for the Lower Indus Plain,

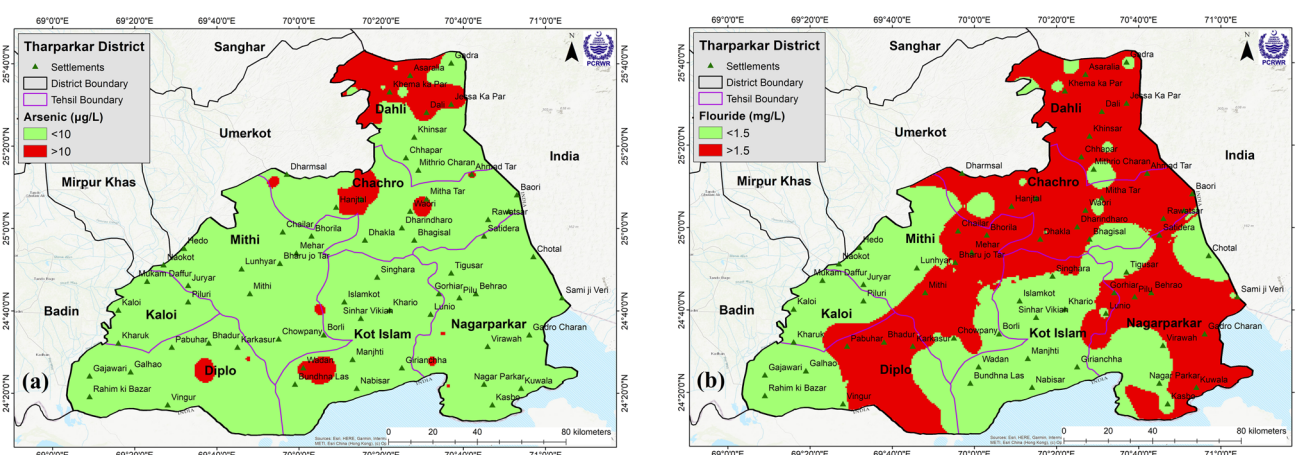


Fig. 17 a) Arsenic and b) fluoride concentrations in the groundwater domain of Tharparkar

which may slightly affect the conversion due to spatial variability in lithology and groundwater chemistry. Site-specific calibration would help improve the accuracy of conversion. Soil sampling was mainly focused on texture analysis, and detailed soil chemical properties were not determined. Therefore, a quantitative assessment of soil–groundwater salinity interactions could not be undertaken. In addition, the study represents a temporal snapshot and does not capture seasonal or long-term variability in recharge and groundwater quality. The development of a hydrogeochemical conceptual model could provide detailed geochemical analyses, which remained beyond the scope of this investigation.

Future research should prioritize site-specific resistivity: EC calibration, integrated soil–groundwater salinity analysis, and long-term monitoring. The integration of ERS with complementary geophysical techniques and numerical modeling would further enhance subsurface characterization and support sustainable groundwater management in hyper-arid environments.

Conclusions

This study provides a regional-scale understanding of groundwater systems in the Thar Desert by integrating high-resolution ERSs with hydrochemical analysis. The groundwater occurrence in the region is highly variable. Shallow groundwater is limited to certain pockets with dominance of deeper aquifers, reflecting low recharge and complex subsurface conditions. Therefore, groundwater availability in the region is not only limited, but also highly unevenly distributed. Moreover, groundwater quality generally worsens with increase in depth, largely due to limited recharge and strong interactions between water and surrounding geological formations. The groundwater quality is strongly influenced by lithology, aquifer depth, and prevailing hydrogeochemical processes, resulting in progressive salinity and fluoride enrichment with depth, while arsenic occurrence remains predominantly geogenic and spatially variable within the sedimentary aquifer system. As a result, deeper aquifers are mostly unsuitable for drinking, while usable groundwater is confined to small, shallow pockets where local conditions are more favorable. The presence of arsenic and fluoride in some areas further complicates safe water access and underscores the need for careful management. Therefore, groundwater alone cannot meet water demands in such a hyper-arid environment without targeted and well-informed strategies. Approaches such as rainwater harvesting, managed aquifer recharge, and selective development of shallow aquifers offer practical pathways to improve water availability. At the same time, the use of saline groundwater for non-drinking purposes, such as saline agriculture, can help improve the livelihoods of communities living in these

regions. Nevertheless, it is imperative to combine geophysical methods with groundwater quality data to better understand subsurface conditions.

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Data availability Data will be made available on reasonable request.

Declarations

Competing interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI use The authors used ChatGPT to improve the language and readability of the manuscript. After using this service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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